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Case Study - Continuous Improvement; Design and Execution of Deepwater Jetting and Drilling-Ahead Operations in the Mediterranean

A. Mohy, S. Elkholy, P. Tan, H. Galanes-Alvarez, and M. Mousa, BP, Cairo, Egypt

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Abstract

This case study focuses on the successful implementation of conductor jetting technique in the West Nile Delta (WND) area of the Mediterranean, highlighting significant improvements in performance, time savings, and cost efficiency. In total, 28 conductors have been jetted successfully by BP in the WND area in the Mediterranean. The first attempt for conductor jetting in BP Egypt was in 2004 and the well was abandoned as the required wellhead stick-up was not achieved. Later in 2007 a successful proof of concept was achieved after successfully jetting 36" conductor. From 2015 to 2018, 20 conductors were jetted as part of a batch-set, top-hole drilling program. The program included jetting the conductor, drilling ahead, landing the surface casing, and temporarily abandoning the well before moving to the next site. After a gap of 7 years; an additional 2 conductors were jetted for wells drilled in Q1 2025. The restart of activities successfully managed the different geological conditions between both locations and maintaining the learning curve in the WND.

A jetting feasibility study was conducted to confirm that soil conditions at the well sites were suitable for jetting. The study concluded that a jetted conductor length of at least 70 meters was feasible. The BP proprietary tool for axial design was used with site specific soil input parameters to generate optimum jetting parameters. The minimum conductor length selected ensured that the surface casing could be landed on the conductors without having to wait on cement or requiring partial landing weights, and with minimum probability of subsidence.

In the first batch setting campaign, 9 conductors were successfully jetted to a target depth of 71.5 meters. Key design and operational learnings were captured and directly applied to the second installation campaign in 2018 and more recently in the last 2 wells drilled in Q1 2025. This systematic approach of lesson learning and procedural updates contributed to a significant optimisation of jetting times, from an initial average of 3.5-5.5 hours in phase 1 (with a maximum 14 hrs. in one well) to 1.5-2.2 hours in phase 2. In the most recent campaign, one well was spud in an existing field and the second was an exploration well in an undeveloped area. During the pre-spud phase studies were carried out to optimize the installation parameters in the first well and to manage the risk in the exploration well and both objectives were successfully achieved. Key operational improvements included the development of detailed Rig Work Instruction that included all learnings and recommendations from previous phases and broader BP worldwide experience

including prescribed weight-on-bit profiles, flow rates, and reciprocation guidance. The selected tricone bit was configured with optimized nozzles and specific bit stick-out requirements were mandated, further enhanced performance. Post-jetting, the 20-inch casings were successfully landed within 33 to 53 hours, and no subsidence or slumping was observed. The BP proprietary tool to estimate theoretical landed weights aligned closely with observed values, validating the design and execution.

In conclusion, the continuous learnings process applied, combined with detailed planning and specialist support onboard the rig during execution, resulted in significant time and cost savings for the WND drilling program. The success of this project demonstrates the potential of the conductor jetting technique to enhance performance in similar offshore drilling environments, particularly when supported by geotechnical data and lessons learned incorporated into operational best practices.

Introduction

Since the initial proof-of-concept trials conducted in 2007, through to the first top hole batch setting campaign executed in 2015, and extending to the most recent operations in 2025, conductor jetting has significantly enhanced well construction efficiency in Egypt's West Nile Delta (WND). Over time, this technique has evolved from a trial approach to a widely adopted standard practice, particularly in areas where geotechnical conditions are suitable. This method has replaced conventional drilling and cementing techniques for conductor installation, offering a more efficient alternative that significantly reduces rig time and overall well construction costs.

The success of conductor jetting in this study is largely attributed to a combination of careful planning and consideration of key environmental and operational factors. These include seabed composition, water depth, and the specific conductor length requirements. Each of these factors plays a vital role in determining the feasibility and effectiveness of jetting operations. During the first top hole batch setting campaign, the absence of advanced simulation tools meant that predictive modeling of jetting parameters in WND was not possible. As a result, the engineering team had to rely on generalized non-site-specific data that were not tailored to the WND site. These data were primarily calibrated using historical performance metrics and operational insights from similar projects in other geographical regions.

Throughout the jetting process, maintaining precise control and implementing robust real-time monitoring proved essential. Key operational parameters such as penetration depth, flow rate, and reciprocation response were continuously tracked and adjusted in real time. This dynamic approach enabled the team to optimise jetting performance real-time, ensuring that the conductor was jetted to the desired depth. Additionally, the final soaking period and subsequent soil-up time were critical in developing adequate skin friction. This friction is necessary to prevent the conductor from subsiding due to its own weight or external loads applied during subsequent phases of well construction.

Following the completion of conductor installation, a series of verification procedures were carried out to confirm the structural integrity of the conductor. This included confirmation of conductor verticality, final penetration depth, and stick-up measurements. Each verification step was to ensure that the installed conductor met all engineering and structural integrity, thereby supporting the long-term integrity and reliability of the well.

WND Jetting Progression

WND Phase 1 (2015)- Building the learning curve

The 2015 phase 1 campaign established critical benchmarks amid significant operational challenges. Initial feasibility studies identified problematic soil conditions including cemented clay layers and sandy intervals below 70m. Based on the available regional geotechnical data and seismic interpretations it was determined

that soil conditions were suitable for the installation of the jetted conductor to minimum depths required that were around ~70m.

However, during execution, one of the wells experienced wellhead subsidence when 180 klbs of casing weight was released. It was also noticed that the rate of penetration below 50 m had reduced to <0.1 m/min in some of the wells, leading to extended jetting time as shown Figure 1

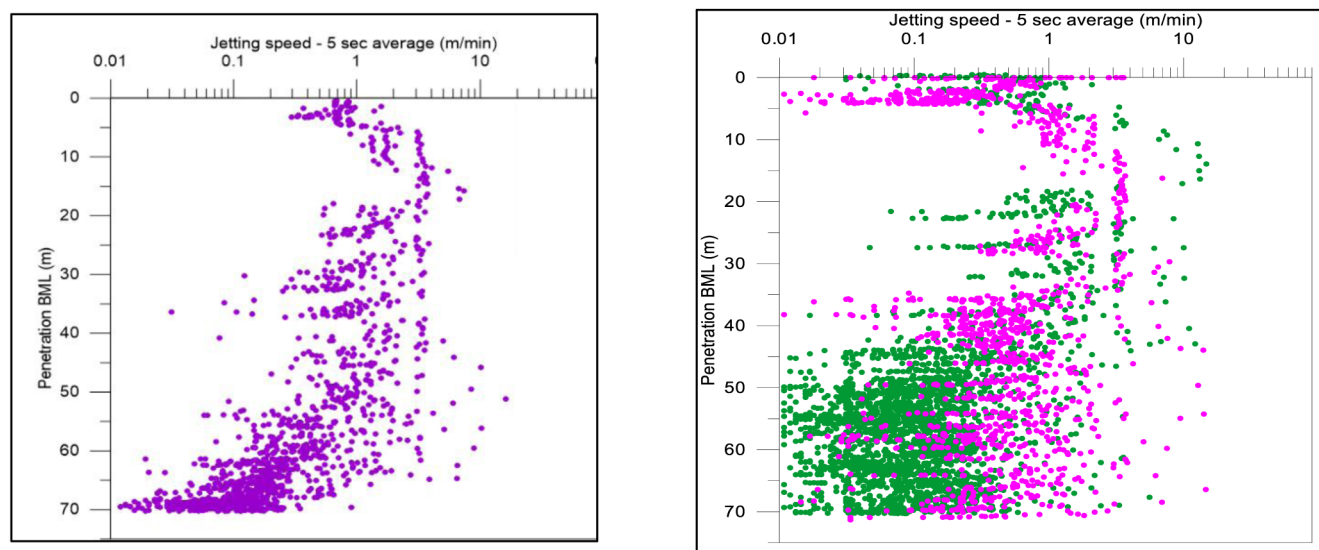


Figure 1—Jetting speed observations - 2015 batch setting campaign

In addition, seabed broaching was observed in one of the wells at 29 m and 52 m penetration when flowrate reached 955 gpm and 1100 gpm respectively, which is believed to be due to exceeding the formation's fracture gradient due to high flow rate. Increased flowrate increased the hydrostatic pressure at the bottom of the hole due to increased friction up the conductor annulus and partial blocking of ports on the low-pressure wellhead housing and drill ahead tool. These incidents led to fundamental changes: modified casing landing sequences (partial set cement incremental release) and staged flow protocols (100 gpm at 5m 700 gpm at 30m 1,100 gpm beyond 50m). Despite these setbacks, the campaign achieved its primary objective: proving jetting viability for top hole batch setting in WND's complex geology.

WND Phase 2 (2017) - Standardization and efficiency

WND phase 2 jetting campaign marked a significant advancement in operational efficiency and execution quality compared to phase 1. Key improvements included a significant reduction in installation times—averaging 1.5 to 2.2 hours per well, nearly 60% faster than the earlier benchmark—achieved through proactive reciprocation strategies that minimized soil setup and maintained high rates of penetration. The campaign also eliminated subsidence during 20inch casing landing, attributed to optimized final weight-on-bit (WOB) and flow rate profiles, adequate soak times, and precise stick-up control. Operational efficiency was further enhanced by minimizing the number of connections and applying lessons learned including earlier initiation of reciprocation, real-time WOB adjustments based on rig measurements, and role specialization between pump control and drilling personnel during critical phases. The use of calibrated depth sensors ensured accurate stick-up measurements even under poor visibility conditions, while pre-connection protocols and contingency planning reduced non-productive time risks. Overall, the campaign validated that continuous improvement through data-driven decision-making, procedural standardization, and adaptive real time decisions can significantly enhance jetting performance while maintaining operational integrity.

WND Phase 3 (2025) - Continuous Improvement

Conductor jetting operations in Egypt's West Nile Delta continued evolving by building on prior lessons to maintain efficiency while ensuring reliability. The journey began with phase 2 breakthrough efficiency, where jetting times averaged just 1.5-2.2 hours—60% faster than previous benchmarks. When one of the wells in phase 3 faced challenging low penetration rates below 50m depth due to unexpected geology, real-time adjustments to parameters and reciprocation length based on previous learning helped regain progress without compromising well integrity. Though this well required record cumulative reciprocations (490m), the real-time feedback provided during drilling operations and adaptability demonstrated that standard procedures could accommodate unexpected soil behavior. No subsidence occurred despite landing WND's heaviest 22x20" casing to date. By the second well, reciprocation timing was further optimized, cutting cumulative strokes to just 56m (lowest in WND) and reducing jetting time to a record 1.14 hours. Once again, casing loads were successfully applied without any displacement being observed.

Key Elements for Jetting Success

Planning:

Detailed planning focusing on geotechnical conditions studies and BP propriety tool analysis is a key for successful jetting operations. These feasibility level studies provided the foundational assessment based on soil parameters like, undrained shear strength, and fracture gradients, confirming jetting viability and identifying likelihood of site-specific risks (e.g., broaching). This was used as an input for BP propriety tool which enabled to optimize conductor make-up and length, establish critical operational recommendations (min/max WOB profiles, reciprocation depths, flow rates..and advised the recommended minimum soil set-up times to ensure axial capacity.

This approach enabled standardized conductor and BHA designs (saving 30% BHA adjustment time), helped generate the flow rate schedule, and advised maximum axial casing loads the soil can withstand. Proper planning helped increase efficiency and mitigate risk and related non-productive time during jetting operations.

Conductor Penetration Length:

Conductor minimum length is mainly determined by axial capacity requirements to support subsequent casing loads, and it is governed by geotechnical parameters (soil strength, set-up time) and operational constraints (top hole section drilling duration). Excessive length could increase jetting time, while insufficient length risked subsidence. Adequate soil data and conductor axial load checks would determine the optimum conductor length. Also, connection design could impact operational efficiency. Accurate measurement of conductor length is also crucial to enable control of the recommended the bit stick-out (2-8 inches). This parameter is also critical for jetting operations. A small error in conductor measurement led to NPT during phase 1 which was mitigated later by supplying BHA spacer subs with wide range of lengths to be able to achieve the required bit stick out within the recommended range and to avoid conductor cutting operations offshore.

BHA and Bit Stick Out:

The configuration of the Bottom Hole Assembly (BHA), particularly the bit stick-out, plays a critical role in achieving efficient and reliable jetting performance. Optimizing bit stick-out (between 2-8 inches) is essential to ensure effective soil erosion at the conductor shoe while minimizing the risk of cuttings pack-off. Deviating from this range could lead to issues such as broaching or slow progress. Bit stick out can be verified using the remotely operated vehicle (ROV) as shown in [Figure 2](#) below. The BHA weight provided adequate weight for penetration and the use of spacer subs was necessary to fine-tune the assembly for varying conductor lengths which achieved the required bit stick-out. This adjustment helped maintain the

neutral point and ensured compliance with the weight-on-bit (WOB) schedule, preventing buckling or accidental release of the drill ahead tool.

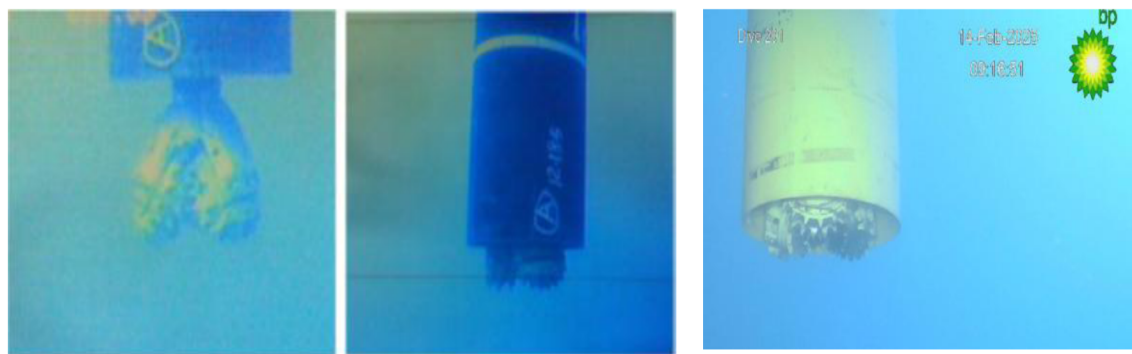


Figure 2—Bit stick out observations from ROV. Image on the left show the correction required after excessive stick-out was observed. Image on the right show typical range of bit-stick out (approx. 4")

Reciprocation:

Reciprocation, defined by its start depth, stroke length, and frequency, proved to be very critical to reduce the WOB to facilitate continuous conductor penetration. The approach evolved significantly throughout the three phases. In phase 1, reciprocation was initiated later than anticipated and conservative short strokes were applied. Maintaining the short strokes was less effective while jetting deeper through more competent formations. Excessive reciprocation in one of the wells resulted in inadequate friction reduction near the target conductor setting depth which may have contributed to conductor subsidence during surface casing landing.

By phase 2, reciprocation strategy was more defined to begin earlier with longer stroke length (if shorter strokes failed to achieve progress) at the deeper competent formation and avoid reciprocations in the last 5m to maintain the WOB within the recommended range and avoid soil damage at TD. This adjustment reduced average jetting times by ~ 40% and helped mitigate subsidence incidents by achieving progress while maintaining soil friction towards the end of jetting operations. Typical ranges of reciprocation lengths for different wells are shown in Figure 3.

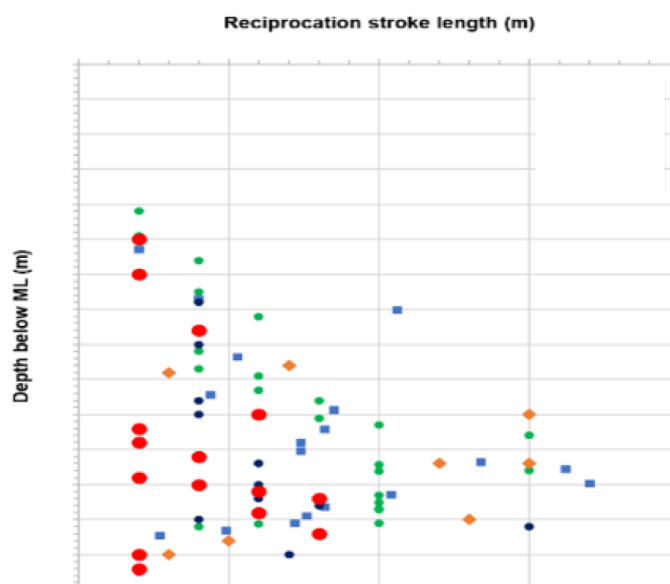


Figure 3—Typical reciprocation lengths in WND

Weight on Bit (WOB) Profile:

The Weight-on-Bit (WOB) profile is a fundamental control parameter in jetting operations and is a function of the soil properties, available weight from wellhead/conductor and the jetting BHA. During phase 1, WOB profiles (Figure 4) generated by BP proprietary tool provided WOB versus depth with a 70% safety factor line to minimise the risk of releasing the drill-ahead tool. These profiles ensured effective jetting while mitigating risks such as releasing the drill ahead tool or subsidence.

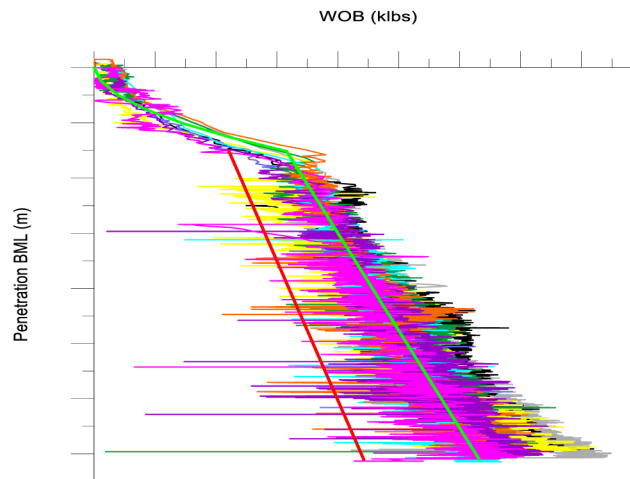


Figure 4—Typical min/max WOB profiles and ranges of m recorded WOB during WND jetting operations

Real-time adherence to the WOB profile proved essential. In WND, usually first reciprocation is required around 22-30 meters, using 10-12-meter strokes to reduce friction and keep the WOB within the specified range. However, failure to maintain the balance between WOB and reciprocation frequency could reduce the friction significantly and allow the WOB to drop below the lower limit which eventually could lead to increasing the conductor soak time (reduce efficiency) or in the worst-case scenario led to subsidence due to significant reduction in friction between the conductor and soil. The WOB profile, when followed, significantly improved efficiency while maintaining structural safety. Jetting outside the profile boundaries introduce the risks of broaching, subsidence, or the accidental release of the drill ahead tool.

Flow Rate:

Flow rate is another critical parameter for jetting operations, directly influencing cuttings transport, efficient mud motor operation, bottomhole pressure management, and broaching risk. Through the phases, flow rate strategy evolved from reactive adjustments to predictive, data-driven strategies. During phase 1, excessive flow rate at shallow depths exceeded the shallow formation's fracture gradient leading to broaching and significant non-productive time (NPT).

In response, phase 2 implemented staged flow schedules, starting at low flow rate at shallow depths then increasing gradually while capping the rates at 1000 gpm below 40m. Strictly adhering to the flow rate schedule prevented broaching in phase 2 and 3.

Performance Benchmarking

Performance metrics show a clear performance improvement (Table 1). Average jetting times decreased from 4.5-14.25 hours (phase 1) to 1.25 hours (phase 3), while reciprocation requirements dropped from 400 m maximum to 56m. Three key performance drivers led to these gains:

- Accurate WOB profile, soaking time and soil set up time based on accurate soil data using BP propriety tool.

- Staged flow rate schedule which helped eliminate broaching.
- Improved reciprocation strategy and frequency based on learnings.

This helped improve the performance and reduce NPT and geotechnical risks (subsidence and broaching).

Table 1—Performance Improvement Across Phases.

Parameter	Phase 1	Phase 2	Phase 3
Jetting Time	4.5-14.25 hrs	1.5-2.2 hrs	1.14hrs (minimum)
Reciprocation (Cum.)	100-400 m	90-280 m	56 m (minimum)
Penetration Rate	0.5-1.2 m/min	1.8-4.0 m/min	3.5-5.0 m/min
Subsidence Incidents	1	0	0
Broaching Events	2	0	0

Operational Challenges

Subsidence

The subsidence of the conductor during the 20-inch casing landing was a critical operational challenge. This happened in one of the wells during phase 1 after landing 20" casing weight (despite allowing 48 hrs for the soil setup) which caused the wellhead to subside, resulting in Non-productive time (NPT). The root cause is believed to be because of inadequate reciprocation strokes length and adhering to shorter strokes when longer strokes were required towards the end. It is believed that the distribution of friction was not regular due to differences in formation properties. In order to progress the conductor to depth, the deeper formation required additional reciprocation that resulted in severe damage to axial capacity and thus did not "set-up" as expected. In that incident, the casing weight was released incrementally after the cement job and waiting on cement. Recommendations for the stroke length and frequency were developed and implemented in later phases, and no subsidence was observed.

Broaching

Broaching is one of the main risks especially while jetting through the shallower unconsolidated formation. During the jetting one of the wells of phase 1, a broaching event was observed at the seafloor by the ROV when the conductor was less than 30 meters of penetration. This incident resulted in significant non-productive time (NPT), primarily due to extended jetting duration and the additional time required for the soil to re-seal around the conductor. Flow rate data show that the flow rate was significantly higher at a shallower depth than in any other well in the campaign. The flow rate reduced to 700 and then 400 gpm in response to the broaching observation. These events occurred prior to the initiation of any reciprocation.

The broaching was temporarily mitigated by reducing the flow rate. However, a second broaching event occurred slightly deeper, using higher flow rate, which was again reduced to control the event. The conductor ultimately reached its target depth and conductor weight was released without signs of subsidence. Nevertheless, during subsequent drilling of the 26-inch hole section, broaching reoccurred after 86 meters of progress. Drilling was stopped for 3.5 hours to allow the soil to stabilize, after which the 20-inch casing was successfully landed 3.4 days post-jetting.

The root cause is believed to be due to using high flow rates at show depths which increased the annular pressure friction and cause the equivalent circulating pressure to exceed the formation fracture pressure. A flowrate schedule was developed based on this incident and was followed strictly. During operations it was also critical to verify standpipe pressures and that circulation was established though wellhead and drill-

ahead tool ports to ensure that no blockages resulted in increased conductor bottom pressures. No broaching incidents occurred after following the flow rate schedule in later phases.

Bit stick out error

Bit stick out is crucial for jetting and requires accurate measure for both the BHA and the conductor. During the jetting of the first well in phase 1, an error in conductor length measurement led to a miscalculated bit space-out. The bit was initially estimated to be positioned within the acceptable range of 2 to 8 inches below the conductor shoe. However, upon assembling the bottom hole assembly (BHA) and conductor, it was discovered that the bit was inside the conductor shoe.

The root cause was traced to a measurement error for one of the 36-inch conductor joints. This resulted in an overestimation of the total conductor length. The issue was resolved by replacing a spacer sub in the BHA with a longer one to achieve the correct bit stick out. While the correction was relatively straightforward, it resulted in 3 hours of rig non-productive time (NPT). The impact could have been more significant had multiple spacer subs or drill collars required replacement. It is recommended that all conductor and BHA measurements be independently verified prior to load-out and again during rig-up. The combination of accurate conductor and BHA tallies minimize the probability of having to carry out conductor cutting operations offshore.

Table 2 below summarises the challenges, corrective action and results:

Table 2—Key Jetting challenges

Challenge	Corrective Action	Result
Subsidence risk	Longer strokes when shorter strokes are ineffective or WOB reduce significantly	No subsidence incidents occurred in later wells/phases
Broaching	Follow WND recommended staged flow schedule	No broaching incidents occurred in later wells/phases
Bit stick-out error	Independently verify all measurements + supply BHA spacer subs with various lengths	Bit stick-out is within range in later wells/phases

Conclusions

In conclusion, the learnings applied, combined with detailed planning and execution, resulted in significant time and cost savings for the WND drilling campaign. The success of this project demonstrates the potential of conductor jetting technique to enhance performance in similar offshore drilling environments, particularly when supported by robust geotechnical data and operational best practices.

The key considerations are as follows:

1. **Geotechnical Feasibility:** The feasibility study is critical to assess soil conditions models, undrained shear strength and potential presence of sand layers or hard stringers, and shallow fracture gradients, confirming jetting viability and identifying site-specific risks (e.g., broaching). This study enables optimization of conductor design and length, to establish critical operational recommendations (WOB profiles, reciprocation depths, flow rates), and advises the recommended soil set-up times for axial capacity prior to casing landing.
2. **WOB Profile and Reciprocation strategy:** Operational procedures include recommendations from the jetting feasibility study, including minimum and maximum allowable WOB against depth of penetration and recommendations for minimum depth for first reciprocation. The number, penetration depth and length of each reciprocation stroke is also carefully managed in conjunction with the Weight on bit recommendations to maximise penetration rate, while minimizing the damage applied to the foundation and the subsequent reduction in axial carrying capacity. It is also critical to record

maximum overpull during reciprocation to ensure that landing string and drillahead tool limits are not exceeded.

3. **Flow rate optimization:** The WND site specific pump or flow rate schedule has been developed based on regional experience and have been adjusted to incorporate operational experience, the effects of usual soil conditions exist. The pump schedule limits the max flow rate at any depth to minimize broaching risk and therefore reduce the probability of low ROP and degradation axial carrying capacity of the soil outside the conductor. The flow rate is also kept above a minimum to avoid packing off inside of the conductor, which in turn can lead to broaching. This risk is particularly relevant when smaller diameter bits (typically 17 'V") are used.
4. **Standardization:** The implementation of standardised conductor make-ups, casing weights/lengths and installation procedures has enabled a significant optimisation of the installation process. Lessons captured during each batch setting campaign have been fed into the operational procedures of the following campaign. This has resulted in measurable improvements of installation time and performance and the reduction of NPT occurrences.

Acknowledgement

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